Generalisation example 2: OpenML car evaluation (categorical, 4-class)

Matric No.: <MATRIC NO> **Name:** <FULL NAME> **Generalisation example** **Variant-2**

Generated by the SKILL `tabular-classification-report` | Model: `Claude Opus 5.5 (claude-opus-5-5)` | Interface: Agent harness - Claude Code 2.1.278 (desktop app) | Code: `https://github.com/BreezyB1n/in6227-assignment1`

Summary. Six ordered categorical attributes, four imbalanced classes (smallest 3.8%), no numeric columns, no test file. Here the comparison is decisive: HGB beats LR by +0.108 macro-F1 (95% CI +0.052 to +0.170), because the target is defined by interacting rules that an additive model cannot express.

1 Data exploration and cleaning

1,382 training / 346 hold-out rows, no missing values or duplicates; the profiler raised only 1 flag.

Table 1: Data issues found by the profiler and the treatment chosen (all statistics learned inside CV folds).

Issue	Evidence	Treatment and rationale
Severe class imbalance	968, 307, 55, 52 training rows; ratio 18.6:1	Primary metric macro-F1; `class_weight` searched; stratified folds.
Only categorical inputs	21 one-hot columns, all levels $\geq 25\%$	LR: one-hot without scaling; HGB: native categorical splits; no pooling needed.

2 Feature selection and engineering

Nothing to engineer: every attribute is a design input. The levels are ordered (low $<$ med $<$ high $<$ vhigh), and a numeric encoding could help a linear model; the skill keeps them nominal because the order of `doors`='5more' and `persons`='more' is not numeric. This choice is worth challenging, and the next ablation to add.

3 Model training

LR (multinomial, one-hot) vs. HGB with native categorical splits; 10-fold CV because $n < 2000$; macro-F1 for refit. HGB's `max_leaf_nodes` first landed on its upper boundary (64), so the range was widened once to 128 per the skill's rule; the re-run selected 66. Stopping: LR converged in 606 iterations; HGB early-stopped after 242 rounds.

Table 2: Hyper-parameter search (randomised, stratified CV, refit on the primary metric) and fitted models. Gap = CV train − validation primary metric (over-fitting indicator).

Model	Search	Selected	Stopping	#inputs	Gap
Logistic Regression	20 cand. over C [0.01, 1000] log, class_weight $\in$ {none, balanced}	C=145, class_weight=balanced	606/5000 iter.	21	+0.034
Hist. Gradient Boosting	30 cand. over learning_rate [0.02, 0.3] log, max_leaf_nodes [8, 128], min_samples_leaf [2, 30], l2_regularization [0.0001, 10] log, class_weight $\in$ {none, balanced}	class_weight=None, l2_regularization=0.0144, learning_rate=0.044, max_leaf_nodes=66, min_samples_leaf=11	242/2000 iter.	6	+0.014

4 Evaluation and comparison

Table 3: Cross-validated and held-out test performance. Class predictions are the arg-max of predicted probabilities. CIs: 1000 test-set bootstrap resamples.

Model	CV f1_macro (mean±sd)	Test Macro-F1 [95% CI]	Bal. Acc.	Accuracy	ROC-AUC (OvR)	Log-loss
Prior baseline	–	0.206	0.250	0.699	0.500	0.838
Logistic Regression	0.896±0.034	0.894 [0.830, 0.948]	0.947	0.945	0.993	0.170
Hist. Gradient Boosting	0.979±0.042	1.000 [1.000, 1.000]	1.000	1.000	1.000	0.005

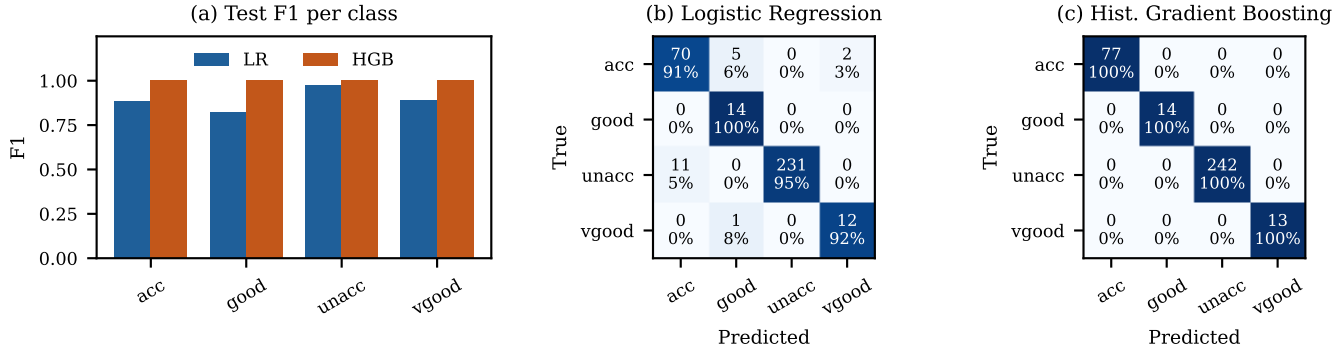

Figure 1: Hold-out set. (a) Per-class F1. (b, c) Confusion matrices (arg-max predictions).

HGB classifies the whole hold-out correctly. LR's errors concentrate where interactions matter: 11 *unacc* cars predicted *acc*. McNemar: 19 rows only HGB gets right, 0 the reverse (p <0.001). Native vs. one-hot encoding for HGB makes no detectable difference here (Table 4).

5 Findings and discussion

Where the main report found a largely additive signal, this dataset is the opposite case, and the same two-model design exposes it: a perfect tree ensemble against a linear model at macro-F1 0.894. The perfect hold-out score should not be over-read: the

Table 4: Ablations: one decision reversed, tuned hyper-parameters kept (so Δ is conservative for the variant). Δ = change in CV f1_macro on the same folds; sd = sd of paired fold differences ($|\Delta| <$ sd: no detectable effect); p: Nadeau–Bengio corrected t-test.

ID	Question	Δ f1_macro	sd	p
A1_hgb_onehot	HGB: one-hot instead of native categorical splits	+0.0070	0.0284	0.605

dataset is a noise-free enumeration of a known rule hierarchy.